



Programming for Data Science

DST314-A,B | Fall, 2026 | 4 Credits

Augsburg University

[NOTE: All course policies are subject to change with notice. - Last updated August 12, 2025]

COURSE INFORMATION

Class Meeting Day and Times: **MWF DST 314 A 12:20 PM - 1:50 PM**

MWF DST 314 B 10:10 PM - 11:40 PM

Class Location: **Hagfors Center 367 - Computer Lab**

Prerequisites, if Any: **CSC165 – Introduction to Computer Programming**
One of MAT 105, MAT 106, or MPL 3

IMPORTANT DATES

- Last day to drop without a 'W' grade & 100% refund - **Wednesday, September 09, 2025**
- Last day to withdraw from all fall classes and receive a 50% refund - **Friday, October 17, 2025**
- Last day to withdraw - Full Semester class - **Friday, November 18, 2025**
- Last day to designate grading option (undergraduate courses) - **Tuesday, December 30, 2025**

INSTRUCTOR INFORMATION

Instructor: **Dr. Sayantica Pattanayak**

Office: Hagfors Center 369G

Office Hours: MF:11:00am – 12:00 pm

W: 2:30 pm-3:30 pm

E-mail: pattanas@augsborg.edu

Preferred Pronouns: She/Her/Hers

COURSE DESCRIPTION

The course is designed for students with a basic Python or programming background and who want to learn the career-building programming skills to succeed in data science and data analytics fields. Topics typically include information manipulation and visualization, text analysis, social network analysis, and machine learning using popular Python toolkits such as Matplotlib, NumPy, pandas, SciPy toolkits (Scikit-learn).classification,regression, Machine learning models. Different metrics for

evaluation. The course includes hands-on exercises throughout the course. The course also includes monthly projects and a final project to demonstrate the new skills.

LEARNING GOALS

At the end of this course, students will be able to:

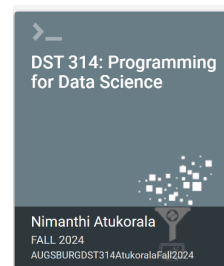
- explain the basic concepts behind the data analysis techniques covered such as data manipulation, visualization, and supervised/unsupervised machine learning
- identify the type of technique preferable to use depending on the type of data analysis.
- perform data analysis and manipulation using data structures and tools provided in the Pandas package.
- use the Matplotlib library of Python for data visualization.
- perform mathematical and scientific computing using the NumPy and SciPy packages.
- gain experience in working with the Scikit-Learn package to perform basic supervised learning machine learning techniques such as classification and clustering.
- apply one or multiple of these techniques learned in small-scale projects simulating real-world applications.
- develop the skills to use data science concepts to improve the efficiency of daily activities.
- develop the confidence to work in a changing working environment.
- be able to install and use other existing or future released (upcoming) data science programming libraries and packages.

TEXTBOOK(S) & COURSE MATERIALS

Required Text(s)

Our recommended course material is zyBook. You can purchase your access code through [zybooks.com](https://www.zybooks.com) or from the bookstore. Once you have your access code, you can join our zyBook course:

1. Sign in or create an account at <https://www.zybooks.com/>
2. Enter zyBook code: **AUGSBURGDST314PattanayakFall2025**
3. Subscribe to zyBook. A subscription is \$64.



Software(s)

IPython: Free and open-source interactive shells. Include browser-based notebook interface.

Google Colab

Anaconda Navigator (<https://www.anaconda.com/products/distribution>)

Other Technology Requirements

In this course, you need to be able to complete the following tasks:

- Check email, access Moodle resources including downloading PDFs, and post PDFs to Moodle.
- Have multiple documents and multiple applications open simultaneously during class: Zoom, Google Jamboards, PDFs, etc.

To complete these tasks,

- You will be using a computer in class. If you do not own one, we encourage you to borrow a computer from the university for the semester.
- You need reliable Internet access for class. We encourage you to find a place to attend a class where you will feel comfortable sharing your video and audio at least some of the time. We understand if that is not possible.
- Please contact your instructor or your faculty mentor/academic advisor if you need help acquiring or using these resources or if you need help finding a place to work where you can comfortably share the video as needed.

Student Access to laptops and the Internet

<https://library.augsburg.edu/equipment>

COURSE POLICIES

Attendance and Participation

Faithful attendance is an absolute requirement for this course. Excessive absences will seriously distract you from both your success and the class as a whole. In addition to being present, you need to engage with the course content and fellow students. Attendance and participation count for 5% of your grade in this course. Important note: arriving late to class or leaving early (more than ten minutes in either case) will count as an absence unless you have notified me at least 24 hours in advance of the situation, and why your late arrival/early departure is necessary.

Here are the behaviors that count:

- arriving on time and prepared
- asking questions
- answering questions
- making comments (extra points for comments that relate to material in the text, and for sharing relevant experiences)

Here are the value-added behaviors—the ones that put your contributions over the top:

- responding to something another student says (including answering a question asked by a student)
- constructively disagreeing with something in the text or said in a class by me or another student

Here are behaviors to avoid:

- coming late and/or leaving early
- inattentiveness and/or disrupting others' learning

- texting or surfing online
- mocking or belittling something said by another person.

If for any reason you are unable to attend class regularly and/or are experiencing difficulty in either keeping up with or understanding the course workload or course structure, please speak with me immediately. It is your responsibility to come to class and know your obligations for all assignments.

Important Note regarding Attendance: Federal law requires faculty to report students who never show up or who stop attending class. This report may cause a recalculation in the student's financial aid award, possibly requiring the student to return funds already received.

Academic Integrity

It is assumed that all students at Augsburg University will conduct themselves ethically and professionally. The [student guide](#), including the [conduct code](#), should guide your behavior.

Academic honesty ([Academic Honesty Policies](#)) is integral to your success in this course and at Augsburg University. The following is an excerpt from the policy:

A University is a community of learners whose relationship relies on trust. Honesty is necessary for the functioning of the Augsburg University community and dishonesty is, therefore, abhorred and prohibited.

One example of how trust is destroyed by a particular form of dishonesty is found in plagiarism and its effects. In its 1990 'Statement of Plagiarism,' the American Association of University Professors (AAUP) Committee B on Professional Ethics notes that one form of academic dishonesty, plagiarism, 'is theft of a special kind [in which] a fraud is committed upon the audience that believes those ideas and words originated with the deceiver. Plagiarism is not limited to the academic community but has, perhaps, its most pernicious effect in that setting. It is the antithesis of the honest labor that characterizes true scholarship and without which mutual trust and respect among scholars is impossible.'

...Penalties imposed by the professor may include a 'zero' or failing grade on the assignment or examination which involved dishonesty, other academic penalties as outlined in the syllabus for the course or other statement of policies distributed by the professor, forced withdrawal from the course, or failure in the course.

When you use someone else's work you must be sure you are not committing plagiarism or violating copyright. If you need assistance with proper citations, you may visit the [Writing Lab](#), [The Gage Center for Student Success](#), or the [Lindell Library](#). I prefer MLA or Chicago Style, however, I will accept APA.

Students who are found in violation of the academic honesty policies concerning plagiarism and copyright infringement will be reported. I reserve the right to lower your grade if you violate the Academic Honesty policy in this course. The student may also face further penalties as determined by the appropriate Dean or university administrator.

Disability Accommodations and Accessibility

Augsburg University is committed to providing an environment where all students have the opportunity to equally participate in the academic experience, including students with disabilities. Students with disabilities have rights as determined by federal and state laws which require

institutions to provide reasonable accommodations for the student's disability to afford an equal opportunity to participate in the university's programs, courses, and activities.

If you need an accommodation because of a disability, please contact The Center for Learning and Adaptive Student Services (CLASS) at 612-330-1053, class@augsborg.edu, or stop by the Gage Center welcome desk on the link level of the Lindell Library to discuss how accommodations may be implemented in your Augsburg courses. If you have a letter from this office indicating you require academic accommodations, please be sure to check in with your instructor to ensure that you get the accommodations you need in this class.

Statement of Inclusion and Participation

Augsburg University values the diversity of persons, perspectives, and convictions. Critical thinking, rigorous analysis, and open discussion across a full range of ideas lie at the heart of the University's mission as an institution of higher learning. Essential to the University in living out its mission of educating students to be "informed citizens, thoughtful stewards, critical thinkers, and responsible leaders" is that the foundation be one of diversity, inclusion, equity, and intercultural competency.

We acknowledge that racism, classism, ableism, homophobia, transphobia, sexism, and other forms of oppression exist. In this class, people of all races, ethnicities, gender expressions and gender identities, religions, ages, sexual orientations, abilities, socioeconomic backgrounds, political affiliations, regions, and nationalities are strongly encouraged to share their rich array of perspectives and experiences. In doing so, we will be asked to take responsibility for our privileges and examine our words and actions. When we know better, we can do better.

Sexual Misconduct, Required Reporting, and Title IX

Augsburg University is committed to combating sexual misconduct. As a result, you should know that University faculty and staff members are required to report any instances of sexual misconduct, including harassment and sexual violence, to the University's Title IX Coordinator so that the student may be provided appropriate resources, support options, and reporting options. What this means is that, as your professor, I am required to report any incidents of sexual misconduct that are directly reported to me, or of which I am somehow made aware.

There are two important exceptions to this requirement:

1. There are confidential resources in the Center for Wellness and Counseling (CWC) and Campus Ministry who do not have this reporting responsibility and can maintain confidentiality. More information can be found on page 15 of the Sexual Misconduct Policy: <http://inside.augsburg.edu/studentaffairs/harassment-and-sexual-harassment-reporting-form/>
2. An important exception to the reporting requirement exists for academic work. Disclosures about sexual misconduct that are shared as part of an academic project, classroom discussion, or course assignment, are not required to be disclosed to the University's Title IX office.

If you have questions or concerns, I encourage you to contact Michael Grewe, Title IX Coordinator, 612-330-1499, or grewe@augsborg.edu

AI POLICY

In all academic work, the ideas and contributions of others must be appropriately acknowledged and work that is presented as original must be, in fact, original. Using an AI-content generator (such as ChatGPT) to complete coursework without proper attribution or authorization is a form of academic dishonesty. If you are unsure about whether something may be plagiarism or academic dishonesty, please contact your instructor to discuss the issue. Faculty, students, and administrative staff all share the responsibility of ensuring the honesty and fairness of the intellectual environment.

In my class, the burden does not fall on the instructor to prove that AI was used, but rather on the student to prove that learning has occurred. If I suspect that you have used AI to complete an assignment instead of engaging with the concepts, your assignment will initially receive a 0, and in order to receive credit for the assignment, you will be required to meet with me to discuss your submission and how you arrived at your response.

Collaboration

You are encouraged to talk to your fellow students about homework problems. Problems can be difficult and collaboration may help you understand the problem and possible solutions. However, when you are writing your solutions to your assignments, you are required to do your own write-ups and propose your own solutions (if group work is permitted, then “your group” must provide your own write-ups and solutions). There are many ways to solve every problem!

Late Work

No late work will be accepted unless prior arrangements have been made with the instructor. The due date and time for each homework are noted on the assignment in Moodle. You may submit your work to Moodle at any time up until the due date for full credit consideration.

COURSE WORKLOAD

- **Participation – in-class individual and group activities, quizzes**

The activities will include points of your attendance as well for the activities. The inclass activities has to be submitted during the class time for full credit.

- **Monthly Projects**

These mini-projects will typically cover the content of the previous two weeks and help you assess your understanding of the material, will improve your programming skills, will help develop your critical thinking skills and your problem-solving skills. A mini-project will normally involve: performing a set of programming-related tasks in Python to analyze a given dataset and answering questions related to your outcome.

- **Final project & the project report:**

A final project will be completed during the last 4-weeks of the semester which provides you an opportunity to demonstrate that you have acquired and sufficiently mastered the topics skills covered in this course. Projects will require you to present your projects to your classmates as well as receive and provide peer review. Submission of a written report is required.

- **Midterm and Final exams:**

Midterm and final exams are designed to allow you to demonstrate that you have acquired and sufficiently understood the terminology and concepts covered in this course. A typical exam will consist of no more than 10 questions.

There will be no makeups for projects and exams

Tentative exam schedule

DESCRIPTION	DATE
Midterm Exam	Thursday, Oct 23, 2025 during the class time

Your Regular Class Schedule	Your Final Exam Schedule
T/H at 10:10 a.m.	18 th Dec,Thursday at 8:30 a.m.
T/H at 12:20 p.m.	16th DecTuesday at 12:00 p.m.

EVALUATION & GRADING

You will be graded on the successful completion of all the exams, assigned work, in-class activities/quizzes, and participation.

DESCRIPTION	PERCENTAGE
Participation – in-class individual and group activities, class post-work, quizzes.	30%
Monthly projects	20%
Final project & report	30%
Midterm exam	10%
Final exam	10%
Total	100%

I use rubrics for major assignments and the Augsburg grade scale as the tentative grade scale.

Grades will be posted in the “Grades” area of our Moodle course. Typically grades will be posted **one week** following the completion of an activity.

If you have questions about the grading, please contact me.

AUGSBURG UNIVERSITY GRADE SCALE			
Grade	Grade Points	Percentage	Description
A	4.00	93-100	Excellent
A-	3.67	90-92.99	
B+	3.33	87-89.99	
B	3.00	83-86.99	Undergraduate: Good Graduate: Satisfactory
B-	2.67	80-82.99	
C+	2.33	77-79.99	
C	2.00	73-76.99	Undergraduate only: Satisfactory
C-	1.67	70-72.99	
D+	1.33	67-69.99	
D	1.00	63-66.99	Poor
D-	0.67	60-62.99	
F	0.00	0-59.99	Failure
P	0.00		Pass*

N	0.00		No Pass
*The letter grade equivalent for a "P" grade is a C-/70 or higher for undergraduate courses and a B/83 or higher for graduate courses.			

Disclaimer: This syllabus is intended to suggest the outline of the course; it is not absolute. Changes to the syllabus will be announced in class and/or displayed on Moodle.